MSc THESIS DEFENCE · DATA SCIENCE

# Synthesis of Mechanistic Phytopathology Models through Composition

## the Simulation of *Tuta absoluta* over Tomato


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University of Verona · Department of Computer Science · 2025–2026



**Reti Innovative Regionali**

Regione del Veneto

PR FESR 2021–2027, Azione 1.1.1 — project AGRIFUTURE




MOTIVATION

# Why do we need a compositional approach for crop–pest models?

01

## What simulation models are for

They reproduce the behaviour of the crop to predict problems before they happen, and they are at the core of agricultural decision support systems.

02

## Crop and pest are two complex systems

Both are mechanistic and multi-stage, each with its own state variables and its own time scales.

03

## Need: reusable composition

Connect existing models through explicit interfaces, so each component stays intact and only the connection is new.

**Goal** make crop–pest simulation easier to adapt, reuse and evaluate.

OUTLINE

# How this work progresses

01 **Problem**
models are hard to reuse

02 **Existing models**
TOMGRO · Milan

03 **Composition**
in PyCrop

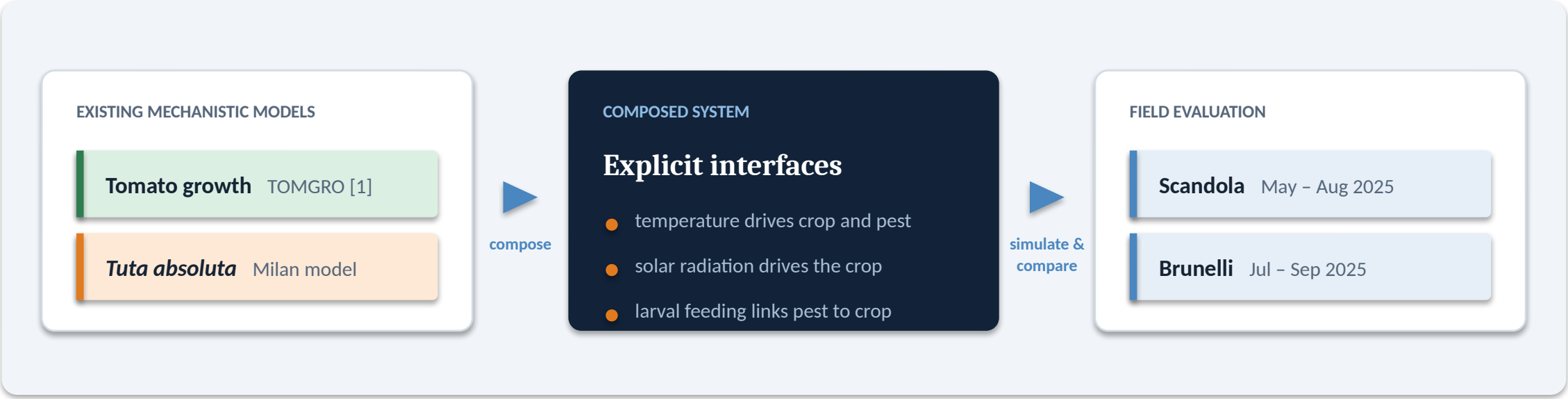
04 **Field data**
Scandola · Brunelli

05 **Evaluation**
against observations

06 **Benchmark**
vs the Milan model

**PyCrop** is a modular simulation framework used to connect and run different models. This work builds the crop–pest composition inside it.

# Objectives of the thesis



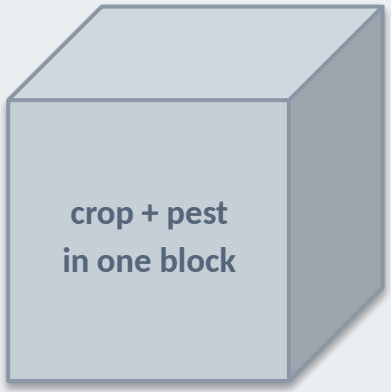

1. Model the tomato–Tuta system under real environmental conditions.
2. Compose existing models through explicit interfaces, preserving their mechanistic structures.
3. Evaluate the composed simulations against field observations and against a reference Milan workflow.

*Evaluated on whether the composed system reproduces the increases, peaks and declines seen in the field, not on absolute abundance.*

[1] Elia Brentarolli et al., A formal approach to crop modeling: Automatic code generation and grid simulation using finite state automata, Computers and Electronics in Agriculture, Vol. 255, 2026.

# From monolithic modelling to compositional modelling

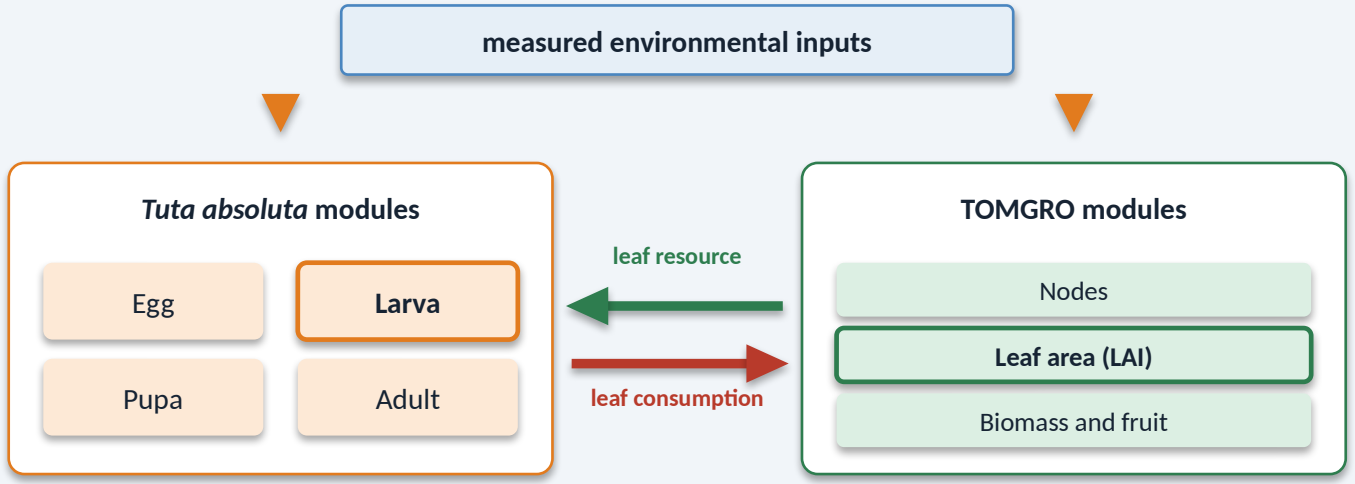
THE USUAL APPROACH




A single formulation. Changing the host or the pest means reworking the whole model.

instead

THIS WORK — COMPOSITION IN PyCrop




*Two existing models, kept whole — only these two variables cross, at every simulation step.*

**One module, one process**

Each module keeps its own equations.

**Bindings carry shared variables**

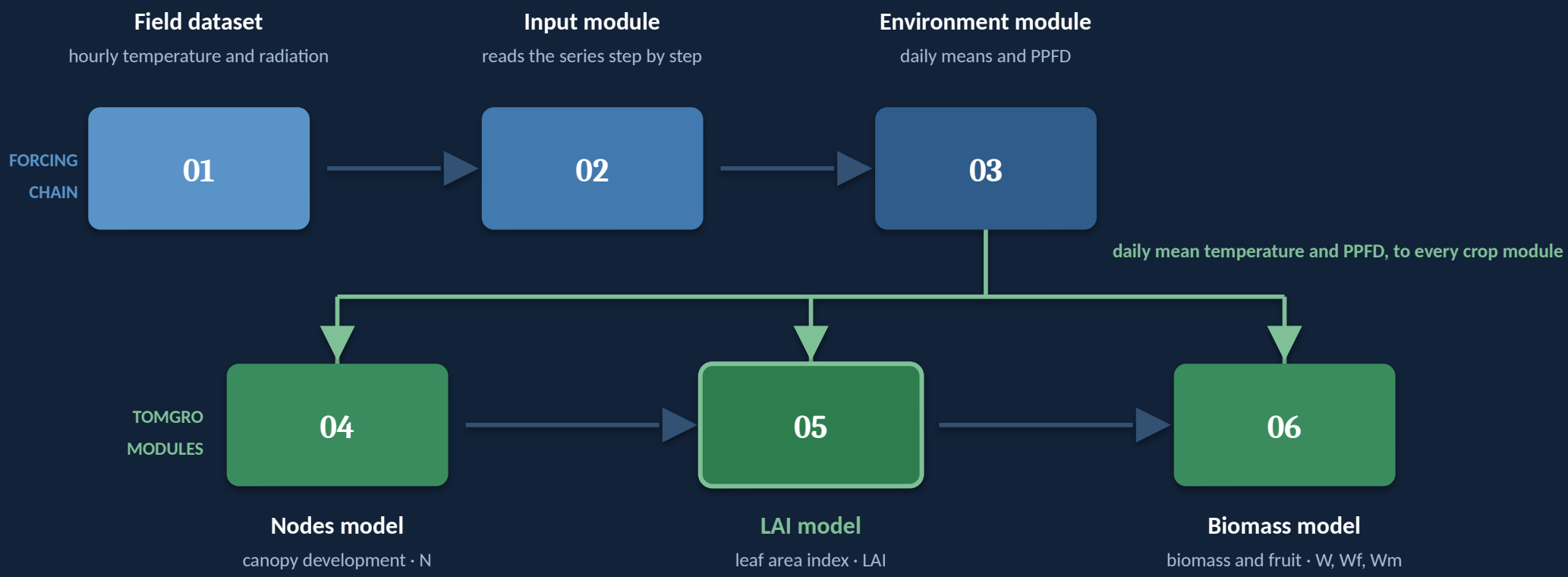
Only what the interaction needs, in agreed units, each step.

**Replace, keep the rest**

A module can be changed without rebuilding the others.

**Key idea** preserve the mechanistic models, but make their interactions explicit and reusable.

# The crop side: how the weather enters, and what TOMGRO carries




**Leaf area index is the state the composition exports.**

*The crop model does not independently represent Tuta absoluta population dynamics.*

# The pest side: the closed *Tuta absoluta* life cycle

STAGE UPDATES AT EACH STEP

**Egg**

$$\Delta E = \varphi(T)A - E/\zeta_e(T) - \mu_e E$$

adults lay eggs · eggs hatch · some die

**Larva**

$$\Delta L = I_l - L/\zeta_l(F) - \mu_l L$$

larvae enter if leaf space allows · pupate · some die

**Pupa**

$$\Delta P = L/\zeta_l(F) - P/\zeta_p(T) - \mu_p P$$

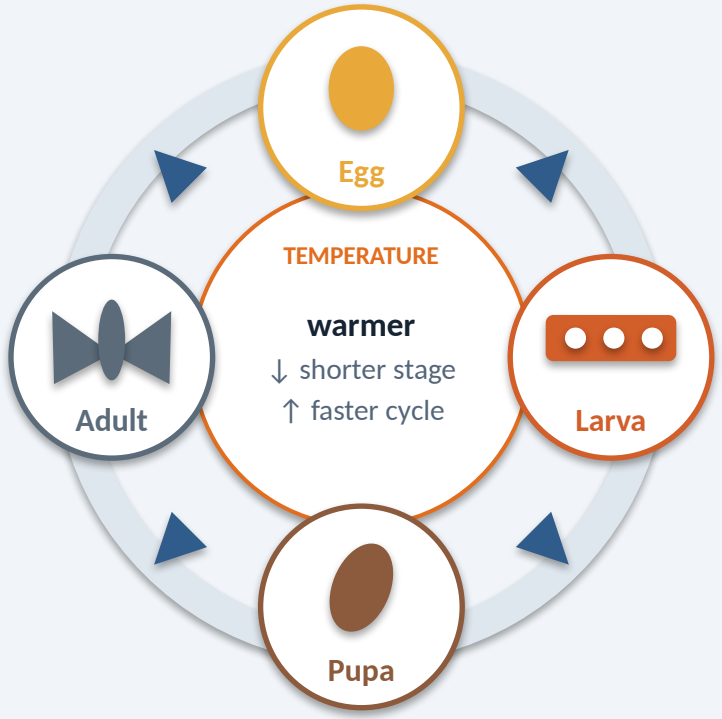
larvae pupate · pupae emerge · some die

**Adult**

$$\Delta A = P/\zeta_p(T) - \mu_a A$$

pupae emerge as adults · adults die

*Every rate depends on temperature; no stage is skipped.*




THE LEAF RESOURCE

## The Milan model carries a simple leaf resource.

One variable with its own growth and a larval consumption term — no canopy, no fruit, no radiation.

**In this thesis it is replaced by the TOMGRO canopy.**

**leaf area → leaf biomass**

PARAMETER SEARCH

For selected parameters I defined a minimum, a maximum and a delta, then used grid search over the combinations to find suitable values.

# The two-way coupling between crop and pest

## 1 CROP → PEST

LAI from TOMGRO

$$C_k = LAI_k \,/\, \nu - L_k$$

available larval space

*Leaf availability sets how many newly hatched larvae can enter.*

## 2 PEST → CROP

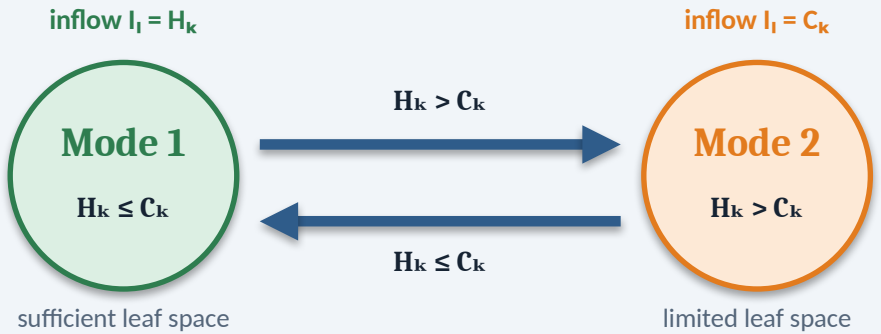
Larval feeding

LAI reduction

Crop growth and fruit biomass

*Larval consumption modifies the canopy, with indirect effects on fruit growth.*

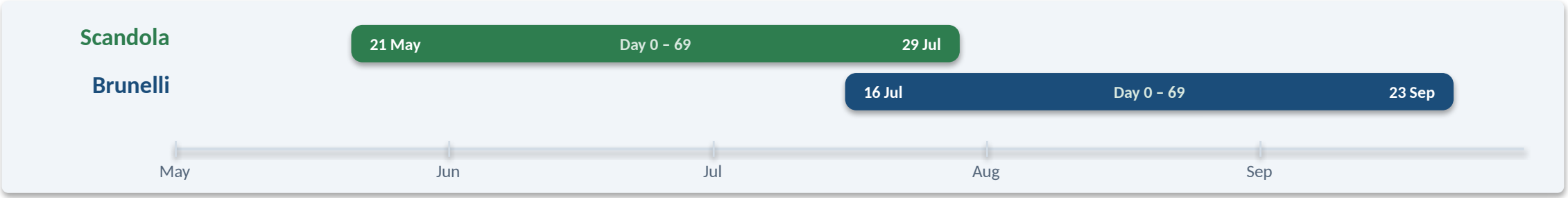
### WHERE THE CROP → PEST DIRECTION ACTS — LARVAL RECRUITMENT SWITCHES BETWEEN TWO MODES




$$I_l = \min(\, H_k \,, \max(0, C_k) \,)$$

- Hatching eggs become larvae only while free leaf space lasts.
- Switching changes the inflow only — the state never jumps.
- $H_k$ hatching eggs in step k · v leaf area needed per larva

# Two monitored tomato crops, in different parts of the season




## Scandola

21 May – 29 Jul 2025

**Role** Staged calibration and first evaluation of the composed model.

*Starts from adults only: no larvae, pupae or mines at the first survey.*

## Brunelli

16 Jul – 23 Sep 2025

**Role** Second field application, and the benchmark against the Milan model.

*All four initial stages estimated with Differential Evolution: eggs and old mines were already present.*

### FROM FIELD OBSERVATION TO MODEL STATE

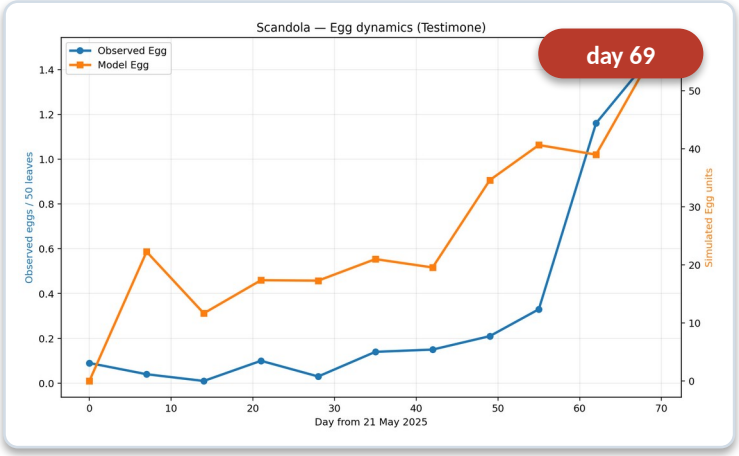
*hourly temperature and solar radiation drive both sites*

- **Field egg counts** → **Egg state, in site-specific units**
- **Larval instars L1–L4, summed** → **one Larva state**
- **Trap catch between inspections** → **Adults, integrated over the same interval**
- **Pupae and empty mines** → context only, not fitted

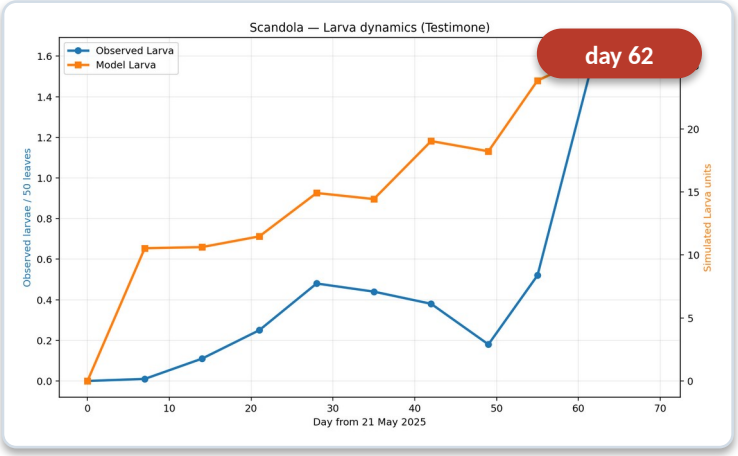
SITE LOCATIONS · SOUTH OF VERONA



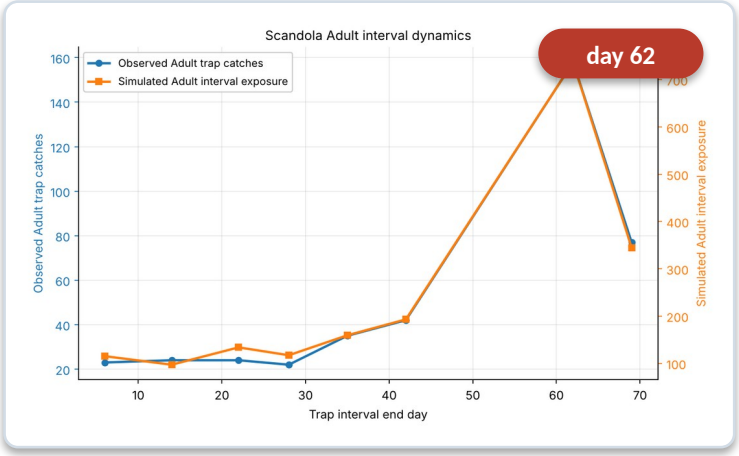
# Comparison w.r.t. Scandola




Eggs




Larvae




Adults, by trap interval

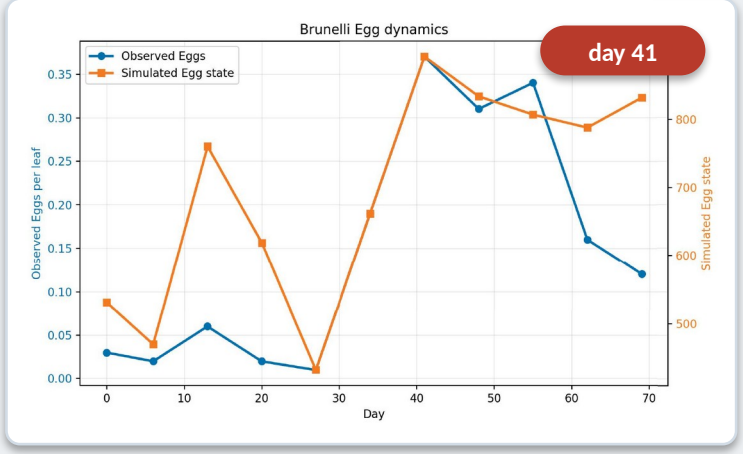
| | Egg | Larva | Adult |
|---|---|---|---|
| Pearson r | **0.806** | **0.780** | **0.998** |
| Direction agreement | **7/10** | **9/10** | **5/7** |
| Peak timing error | **0 d** | **0 d** | **0 d** |

**READING THE ADULT COLUMN**

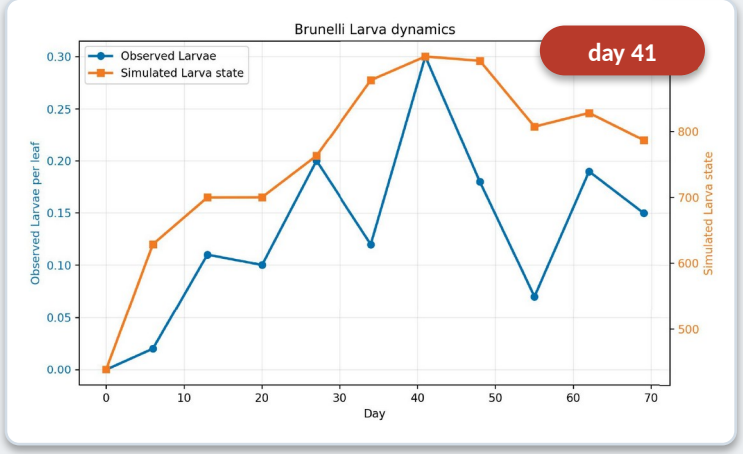
- Trap catches accumulate between inspections, so adults are integrated over each interval.
- Intervals run from 6 to 20 days, and the largest catch falls in the single 20-day one.
- Eight trapping intervals of unequal length provide seven direction comparisons.

RESULTS

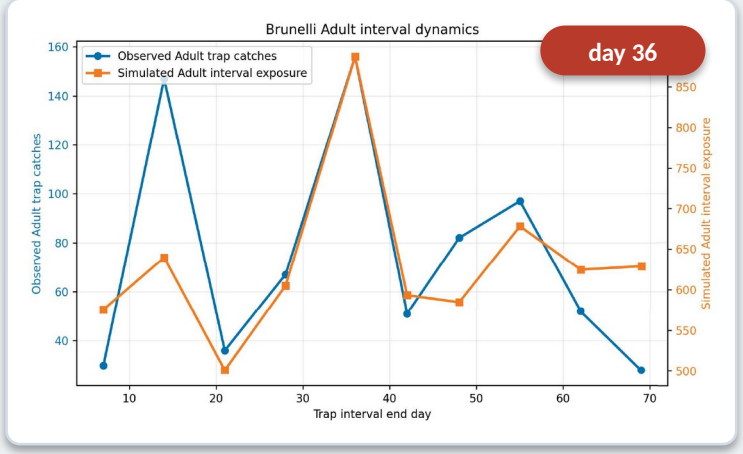
# Comparison w.r.t. Brunelli




Eggs




Larvae




Adults, by trap interval

| | Egg | Larva | Adult |
|---|---|---|---|
| Pearson r | **0.784** | **0.778** | **0.739** |
| Direction agreement | **8/10** | **8/10** | **7/9** |
| Peak timing error | **0 d** | **0 d** | **0 d** |

**WHAT DIFFERS FROM SCANDOLA**

- The infestation had already started, so all four initial stages were estimated rather than set to zero.
- Ten trapping intervals here, against seven at Scandola.
- Same evaluation criteria, although the two sampling calendars are not identical.

CROSS-SITE COMPARISON

# Same architecture, two very different field situations

| | Scandola | Brunelli |
|---|---|---|
| Egg Pearson r | 0.806 | 0.784 |
| Larva Pearson r | 0.780 | 0.778 |
| Egg direction | 7/10 | 8/10 |
| Larva direction | 9/10 | 8/10 |
| **Peak timing error** | 0 d | 0 d |
| **Leaf area lost by day 69** | 3.0 % | 25.8 % |
| **Fruit biomass lost** | 1.1 % | 12.7 % |

*Adults are not compared across sites: the two records have a different number of trapping intervals.*

HOW EACH RUN STARTED

**Scandola** adults only

No larvae, pupae or mines at the first survey, so only an adult state was set.

**Brunelli** all four stages

The infestation had already begun, so the starting state was estimated by Differential Evolution.

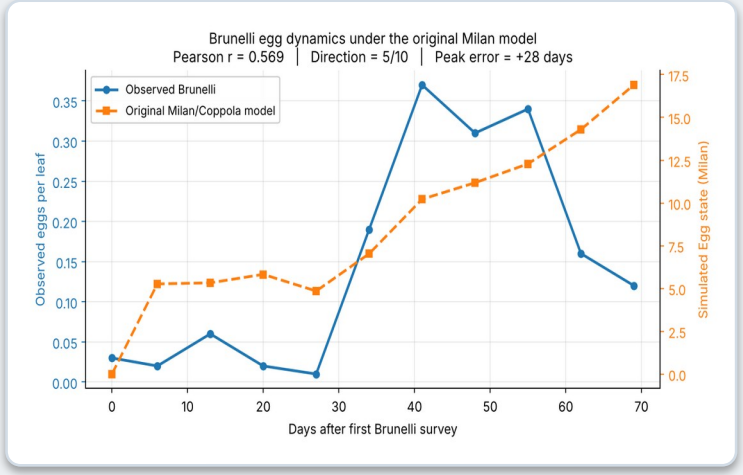
HOW LEAF DAMAGE AFFECTS SIMULATED FRUIT BIOMASS

*model-predicted effect, not a measured yield loss*

| | | |
|---|---|---|
| **Larval feeding** | | |
| **LAI reduction** | −3.0 % | −25.8 % |
| **Simulated fruit biomass** | −1.1 % | −12.7 % |

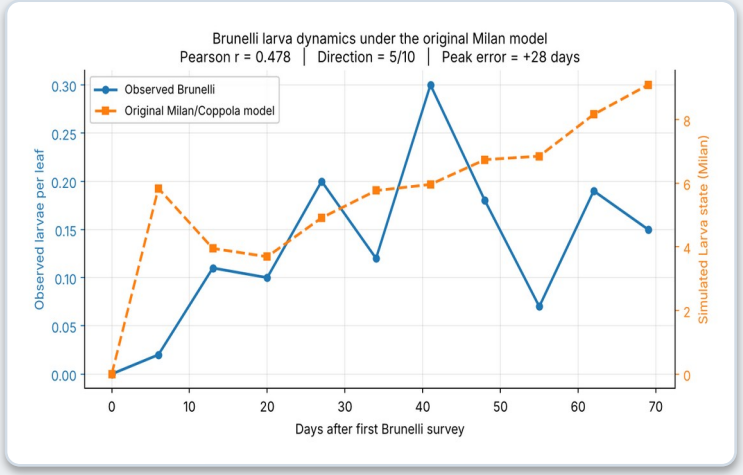
**Each site was calibrated separately.** So this is not a transfer test with unchanged parameters, and the gap between 3.0 % and 25.8 % cannot be attributed to climate alone — infestation history and site-specific damage settings differ too.

# The original Milan model keeps rising after the field has peaked



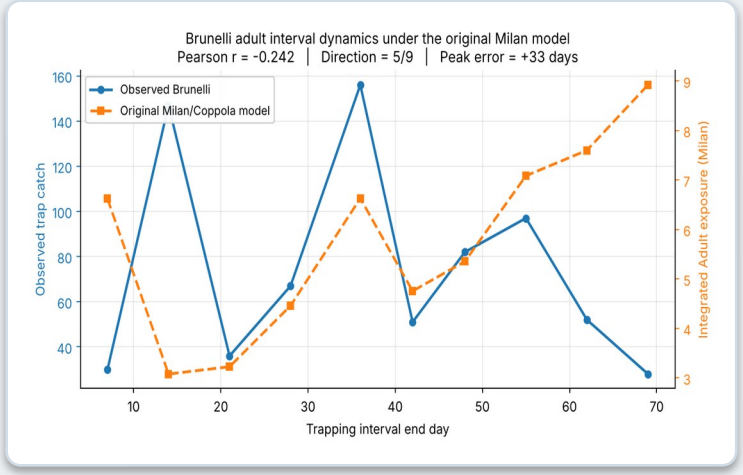

**Eggs**

peak 28 d late




**Larvae**

peak 28 d late




**Adults, by trap interval**

peak 33 d late

**An independent run, not a re-fit.**

The Milan reference was independently reconstructed from its source biological parameters together with documented crop-resource assumptions, and run on the Brunelli weather. No Brunelli observation was used to recalibrate it; both models were then scored on the same survey dates with the same definitions.

# The composed workflow gains at every stage

Same Brunelli observations, same survey dates, same scoring: **original Milan model** → **final composed workflow**

**+0.22**

**Egg** correlation

0.569 → 0.784

**+0.30**

**Larva** correlation

0.478 → 0.778

**+0.98**

**Adult** correlation

−0.242 → 0.739

**28–33**

**days** of delay removed

+28, +28, +33 d → 0 d

| Egg direction agreement | Larva direction agreement | Adult direction agreement | Dominant egg and larva peak |
|---|---|---|---|
| 5/10 → **8/10** | 5/10 → **8/10** | 5/9 → **7/9** | day 69 → **day 41** |

**A system-level comparison.** The two configurations differ in more than the crop representation, so the gain belongs to the whole workflow, not to TOMGRO alone.

CONCLUSIONS

# What the work achieved, and where it goes next

## WHAT WE ACHIEVED

### 1 Model composition

TOMGRO and the existing Tuta absoluta mechanisms were combined through explicit PyCrop bindings.

### 2 Field evaluation

The dominant egg and larva peak timing was reproduced at both Scandola and Brunelli.

### 3 Improved temporal agreement

On the Brunelli window the final workflow agreed with the observations more closely than the Milan reference.

## FUTURE WORK

### 1 Observation model

Link the simulated pest states to measured leaf counts and trap catches, so amplitude can be evaluated as well as timing.

### 2 Transferability

Test the framework across new seasons and greenhouses with less recalibration.

### 3 Towards a digital twin

Extend the damage modelling and introduce real-time data assimilation.

DISSEMINATION

# The work has been presented beyond the thesis



## Scrittura facilitata di modelli predittivi mediante tecniche di composizione: il caso Tuta absoluta su Pomodoro


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*Questa ricerca è stata finanziata da Regione del Veneto PR FESR 2021-2027 Azione 1.1.1 nell'ambito del progetto "AGRIFUTURE: IL FUTURO DELLA SOSTENIBILITÀ PER LE SFIDE COMPETITIVE DELLE AZIENDE AGROALIMENTARI VENETE" (ID 24279_001587_04499230235)*


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Bari, 10–11 September 2026


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*Funded by Regione del Veneto, PR FESR 2021–2027, Azione 1.1.1 — project AGRIFUTURE.*

# Thank you

**Ali Bahar**

Supervisor Prof. Davide Quaglia Co-supervisor Dott. Elia Brentarolli

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Reti Innovative Regionali, Regione del Veneto · PR FESR 2021–2027, Azione 1.1.1 · project AGRIFUTURE

# How the parameters were set

### A Published values first

Every parameter that the Milan formulation and TOMGRO already provide was taken as published and not re-estimated. This fixes most of the model before any search begins.

### B A bounded search for the rest

The few remaining parameters were set with the MIN / MAX / DELTA grid protocol: a declared range and step for each one, every combination run, and the outcome read from the whole grid rather than from a single hand-chosen run.

### C Initial states per site

Scandola began from adults only, since the first survey found no larvae, pupae or mines. At Brunelli the infestation was already under way, so the four initial stage values were estimated by Differential Evolution.

**KEPT OUT OF THE FIT**

- The evaluation series were never normalised or rescaled.
- No per-stage scaling coefficient was fitted to close the unit gap.
- Metrics and the comparison window were fixed before the results were read.

*Calibration is per site: the two farms share the architecture, not a single fitted parameter set.*